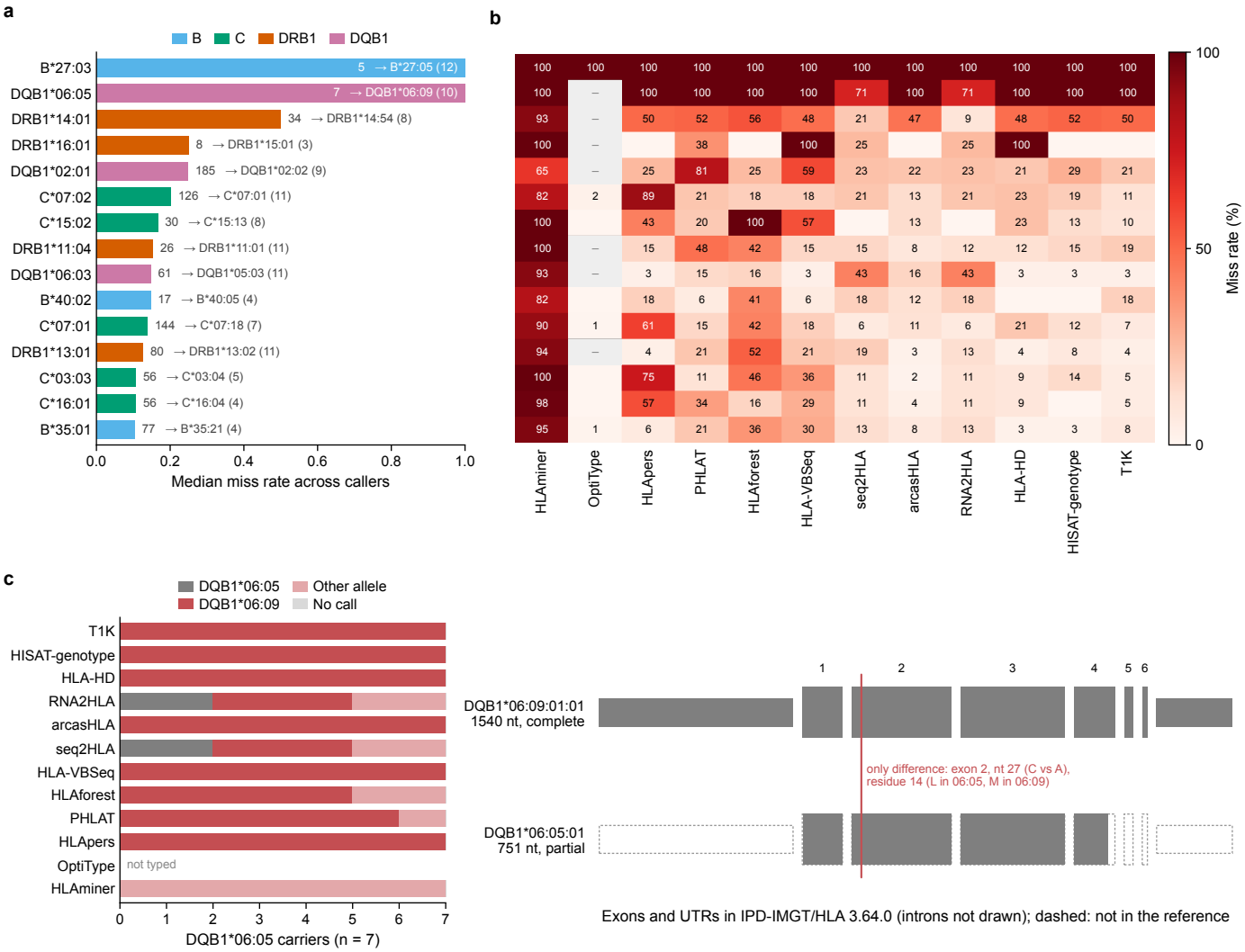
a
B
C
DRB1
DQB1
B*27:03
DQB1*06:05
DRB1*14:01
DRB1*16:01
DQB1*02:01
C*07:02
C*15:02
DRB1*11:04
DQB1*06:03
B*40:02
C*07:01
DRB1*13:01
C*03:03
C*16:01
B*35:01
5 → B*27:05 (12)
7 → DQB1*06:09 (10)
34 → DRB1*14:54 (8)
8 → DRB1*15:01 (3)
185 → DQB1*02:02 (9)
126 → C*07:01 (11)
30 → C*15:13 (8)
26 → DRB1*11:01 (11)
61 → DQB1*05:03 (11)
17 → B*40:05 (4)
144 → C*07:18 (7)
80 → DRB1*13:02 (11)
56 → C*03:04 (5)
56 → C*16:04 (4)
77 → B*35:21 (4)
0.0
0.2
0.4
0.6
0.8
1.0
Median miss rate across callers
b
HLAminer
OptiType
HLApers
PHLAT
HLAforest
HLA-VBSeq
seq2HLA
arcasHLA
RNA2HLA
HLA-HD
HISAT-genotype
T1K
Miss rate (%)
100
50
0
c
DQB1*06:05
DQB1*06:09
Other allele
No call
T1K
HISAT-genotype
HLA-HD
RNA2HLA
arcasHLA
seq2HLA
HLA-VBSeq
HLAforest
PHLAT
HLApers
OptiType
not typed
HLAminer
0
1
2
3
4
5
6
7
DQB1*06:05 carriers (n = 7)
1
2
3
4
5
6
DQB1*06:09:01:01
1540 nt, complete
only difference: exon 2, nt 27 (C vs A),
residue 14 (L in 06:05, M in 06:09)
DQB1*06:05:01
751 nt, partial
Exons and UTRs in IPD-IMGT/HLA 3.64.0 (introns not drawn); dashed: not in the reference